

Objective Questions

Q.1 What is multicollinearity?

- a) The presence of multiple dependent variables
- ☒ b) The presence of correlated independent variables
- c) The presence of missing data in the dataset
- d) The presence of outliers in the dataset

Q.2 In a survey comparing the preferred modes of transportation to work among residents in different neighborhoods of a city, which statistical test would be most appropriate for analyzing the differences in transportation preferences?

- a) Paired t-test
- b) ANOVA
- c) Pearson correlation coefficient
- ☒ d) Chi-square test

[Assuming Non parametric data]

Q.3 The Benjamini-Hochberg correction is designed to:

- a) control the family-wise error rate
- b) control the false positive rate
- ☒ c) control the false discovery rate
- d) increase the power of a statistical test

Q.4 What is the main drawback of using correction procedures to address the problem of multiple comparisons?

- a) They increase the likelihood of false positives
- b) They decrease the likelihood of false positives
- ☒ c) They increase the likelihood of false negatives
- d) They decrease the likelihood of false negatives

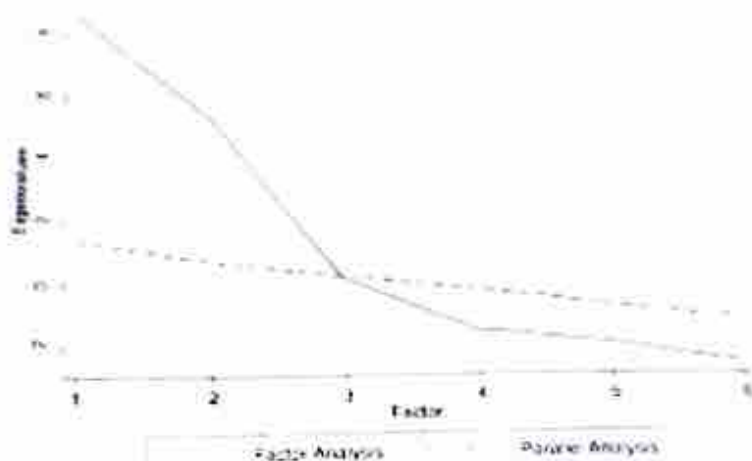
Q.5 What is the problem of multiple comparisons?

- a) It is the issue of comparing one sample to multiple populations
- b) It is the issue of comparing multiple samples to one population
- ☒ c) It is the issue of testing multiple hypotheses simultaneously
- d) It is the issue of testing a hypothesis without a control group

Q.6 Why is ANOVA preferred over multiple t-tests? Check all that apply

- ☒ a) Compares all means simultaneously
- ☒ b) Reduces type 1 error
- c) Increases chances of finding a significant effect
- ☒ d) Balances within-subject error

Q. 14 Based on the following plot, pick all the statements that are true



- ☒ a) Eigen-Value criterion **overestimates** intrinsic dimensionality
- ☐ b) Eigen-Value criterion reveals an intrinsic dimensionality of 2
- ☒ c) **Parallel analysis reveals an intrinsic dimensionality of 2**
- ☐ d) None of the above is true

Q. 15 A new fitness instructor at IIIT is interested in finding the best fitness method for UG students. He conducts weight training exercises in one batch of students and running exercises in another batch of students over a 3 month period with fitness levels measured every month. Which is the best statistical test for analyzing such data?

- ☐ a) 2x3 ANOVA
- ☐ b) Multiple ANOVA
- ☒ c) Mixed ANOVA
- ☐ d) Paired t-tests

Q. 16 A researcher wants to determine if there is a difference in the mean scores of students in three different teaching methods (lecture, group discussion, and hands-on activities). Which of the following statistical tests is appropriate for analyzing this relationship?

- ☐ a) t-test
- ☒ b) ANOVA
- ☐ c) Pearson correlation coefficient
- ☐ d) Chi Square test

now wants to create a sampling distribution of the mean proportion using simulations. What is the right way to do this?

- a) Randomly draw 10,000 values from a $\text{Normal}(0.3, 0.1)$, and plot the distribution and compute the mean of this distribution. The answer will be very close to 0.3 that was observed as the mean of the real data sample, and therefore is the closest approximation to the sampling distribution of the mean.
- ✓ b) Randomly draw 100 values from a $\text{Normal}(0.3, 0.1)$, compute the mean of this sample, and repeat 10,000 times to get and plot a distribution of 10,000 means. This is called the sampling distribution of the mean, given $N = 100$.
- c) Randomly draw 100 values from a $\text{Normal}(0.3, 0.1)$, compute the standard deviation of this sample of 100 values, and simulate this process 10,000 times to get the mean of all such standard deviations. Therefore, the sampling distribution would be a Normal distribution with mean = 0.3 and standard deviation calculated from the mean of 10,000 simulated samples.

Q. 13 Akash measures people's preference for chocolate brand A based on a survey sent out to 500 people and reports that 68% of people like brand A with a 95% confidence interval of [55%, 81%]. This means that:

- a) Akash is 95% confident that the true percentage of all people who like brand A lies in the interval of 55% - 81%.
- b) If Akash were to repeat the measurement many times, he would expect to find that around 68% of the people preferred brand A in about 95% of future measurements.
- c) Akash is 95% confident about 68% being the true percentage of all people who like brand A.
- ✓ d) If Akash were to construct confidence intervals using a similar procedure in the future, he believes that 95% of those confidence intervals will contain the true percentage of all people who like brand A.

Q. 7 In a study investigating the association between hours of sunlight exposure and vitamin D levels in individuals, which statistical test would be most appropriate for analyzing this relationship?

- a) t-test
- b) ANOVA
- ☒ c) Pearson correlation coefficient
- d) Chi Square test

Q. 8 A pharmaceutical company recruits two groups of people to test their new drug. One group is given the drug and the other receives a placebo treatment. What test should you use here?

- a) Paired t test
- ☒ b) Independent t test
- c) Anova
- d) Student's t test

[Assuming: parametric data]

Q. 9 Given a set of candidate models for a dataset, the AIC values for the models are -237.84, 3.54, -201.92 and 569.54. The preferred model would be the model with AIC value equal to

- ☒ a) -237.84
- b) 3.54
- c) -201.92
- d) 569.54

[Assuming: Model performance decreases with increase in AIC]

Q. 10 Which of the following correction procedures is commonly used to address the problem of multiple comparisons?

- a) The t-test
- b) The chi-squared test
- ☒ c) The Bonferroni correction
- d) The ANOVA test

Q. 11 Billu is one of the new cats at IIIT-H. Students have noticed that Billu seems to prefer hanging out in the Parijat hostel rather than in OBH Hostel. To confirm if this behavior is consistent among other cats in IIIT-H, they decide to observe 50 cats' preferences. What statistical test could they use to analyze this categorical data? (Hint: assume that the expected distribution of cats across the two hostels should be equal)

- ☒ a) Chi-Square Test for Goodness-of-Fit
- ☒ b) Two Sample related t-test
- c) The Binomial Sign Test
- d) One-way ANOVA

[Assuming: Non-param data]

Q. 12 An exit poll based on a data sample of 100 people suggests that 30% of the population is likely to vote for XYZ party. A statistician makes an assumption that the population distribution (of this proportion) is a Normal distribution with mean = 0.3 and standard deviation = 0.1. He

Q.3 You are conducting a study on the factors that influence the job satisfaction of employees in a large organization. The variables you are analyzing include job security, work-life balance, compensation, job autonomy, interpersonal relationships with colleagues, opportunities for advancement, and the level of challenge in the job. You believe that these variables are interrelated and that there may be underlying factors that contribute to job satisfaction.

What would be your proposed methodology to check for such underlying factors? (1 mark)

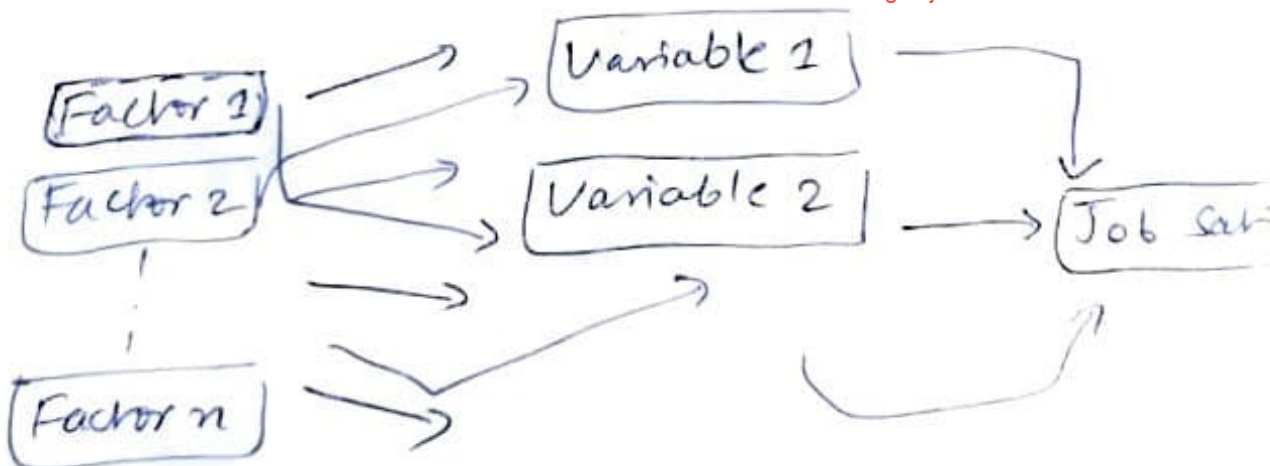
⇒ Factor Analysis would be the proposed methodology to check for such underlying factors as it could capture the latent factors/variables that actually contribute to deciding variables impacting Job satisfaction.

⇒ After performing Factor Analysis, Factor loadings & Factor scores are calculated signifying the proportion of variance explained for each variable [like job-security].

These factors can be found using a priori/latent information & Factor rotation can be applied to distribute variance to get meaningful variance for each factor.

Before performing Factor analysis, we can perform method of sphericity correction that checks if rows in correlation matrix are orthogonal.

Correct Answer: Factor Analysis should be used, because it helps identify underlying latent factors among several interrelated variables contributing to job satisfaction.



You are conducting a study on factors that influence job satisfaction among employees in a large organization. The variables you are analyzing include job security, work-life balance, compensation, job autonomy, interpersonal relationships with colleagues, opportunities for advancement, and the level of challenge in the job. You recognize that these variables are interrelated and that there may be underlying factors contributing to job satisfaction. What methodology would you propose to investigate these underlying factors?

Q. 17 What does a high value of Cook's distance indicate in the context of regression diagnostics?

- a) A predictor variable is perfectly collinear with another predictor
- b) The model perfectly fits the data without any outliers
- ✓ c) A data point might be an influential outlier affecting the regression coefficients significantly
- d) The residuals are normally distributed

Descriptive Questions:

Q. 1 Twenty people each perform a sensorimotor task in two conditions, one in a quiet room, alone, the other is a brightly lit room with a dozen people watching. An electronic timer takes an accurate record of the number of errors made in each condition.

- (a) What test would be appropriate for investigating the significance of the differences in performance between the two conditions? (0.5 mark)
- (b) Assume everybody deteriorates in the second condition. What test would be appropriate for seeing whether individuals tend to deteriorate to the same degree? (0.5 mark)

One-sample t-test on the differences

A) Paired t-test

Assuming that each participant undergoes experiments in both conditions, we can calculate $t\text{-value} = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{s_1^2/n_1 + s_2^2/n_2}}$ and comment on the difference based on significance level & rejecting/accepting Null hypothesis

B) Independent t-test / Student's t-test

Since now we only have one group to find significance of differences in, we can formulate Null hypothesis &

H_0 : All participants deteriorate to same degree..

H_1 : All participants don't deteriorate to same degree
there are differences

By comparing obtained values with level of significance we can come to a conclusion

Q. 2 Does drinking coffee in the evening impair sleep more than drinking tea in the evening?

State the following. (1 mark)

- A) Null hypothesis
- B) Alternate hypothesis
- C) Independent variable & Dependent variable
- D) Statistical test

⇒ A) Drinking coffee in the evening impairs sleep rate in similar factor of impairment as drinking tea in the evening: H_0

⇒ B) Drinking coffee in the evening impairs sleep more than drinking tea in evening: H_1

Drinking coffee in the evening impairs sleep less than drinking tea in the evening: H_2

⇒ C) Independent variables: Amount of coffee consumed in the evening & Amount of Tea consumed in the evening

Dependent variables: Degree of sleep impairment

⇒ D) Statistical test: Multiple Regression

$$Y = \sum_{i=1}^n b_i X_i + b_0 + \epsilon \quad [n=2]$$

Here after getting values of b_i we can compare which predictor has more impact on sleep deprivation based on the magnitude/value of b_i

A) Drinking coffee in the evening does not impair sleep more than drinking tea in the evening.

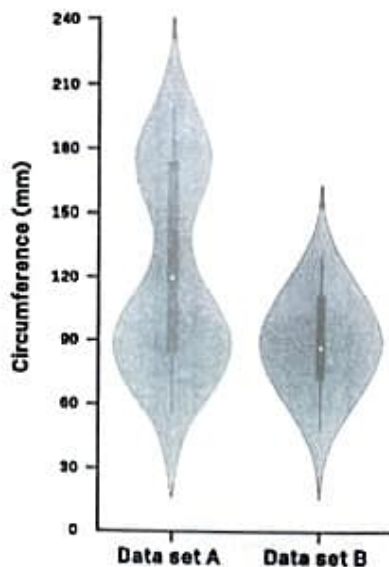
B) Drinking coffee in the evening impairs sleep more than drinking tea in the evening.

C) IV: Type of drink consumed in the evening.

DV: Level / degree of sleep impairment

D) Independent samples t-test

(one-tailed, assuming parametric data)



i) Who showed more consistency in the size of the cupcakes, according to the central tendency and spread indicated in the violin plots?

- a. Contestant A
- ☒ b. Contestant B
- c. Both showed equal consistency
- d. It is not clear from the plots

B

ii) What does the presence of two distinct peaks in Contestant A's data suggest?

- a. Contestant A made two different types of cupcakes.
- ☒ b. Contestant A's cupcakes varied greatly in size.
- c. The measurement of Contestant A's cupcakes was inaccurate.
- d. Contestant A has a very consistent baking size with minor variations.

B

7. In a meticulous review of warfare tactics in the Seven Kingdoms, two distinct approaches were employed by renowned commanders: Strategy A, favoured by Tywin Lannister, and Strategy B, preferred by Robb Stark. Overall data initially suggested that Tywin's Strategy A led to more triumphs than Robb Stark's Strategy B. However, upon closer examination of the battle outcomes based on different terrains—such as the icy landscapes of Beyond the Wall, the rocky terrain of The Eyrie, or the dense forests of the North—it was observed that Robb Stark's Strategy B consistently outperformed Tywin Lannister's Strategy A in each specific terrain type.

Which statistical phenomenon is most likely in play here?

- ☒ a. Simpson's Paradox
- b. Selection bias
- c. Experimenter Bias
- d. Regression to mean

Assumption:

When in each type / department, another section outperforms as compared with the cumulative one ~~which is taken~~

A

- ☒ c. $146/\sqrt{50}$ cm
d. 148 cm

C

4) To make the guess above, they had to rely on the central limit theorem (CLT). Which of the following statements (pick more than 1 if appropriate) are true of the CLT? For a sufficiently large sample size, [1 mark]

[CREDIT ONLY IF ALL CORRECT ANSWERS ARE PICKED]

- ☒ a. the sample mean is equal to the population mean
☐ b. the sample standard deviation is equal to the population standard deviation
☒ c. the distribution of the sample means can be assumed to be a normal distribution
d. the standard deviation of the sampling distribution of the mean is small

A, C

5. A meteorologist is predicting the likelihood of rain tomorrow. He has historical data but also wants to incorporate his beliefs based on recent weather patterns and personal expertise. Which approach to probability allows incorporating such subjective beliefs along with the historical data?

- a. Frequentist Probability
☒ b. Bayesian Probability
c. Empirical Probability
d. Classical Probability

B

6. Two-part Question

In a local baking competition, two contestants were asked to bake cupcakes with the goal of achieving a specific circumference for each cupcake to ensure uniformity, which is a criterion for judging. The following violin plots illustrate the distribution of the circumferences of cupcakes baked by each contestant, where Data set A shows the circumference of Contestant A's cupcakes, while Data set B shows the circumference of Contestant B's cupcakes. Based on this, answer the following questions:

Objective Questions (7.5 marks):

1. Anna conducts a survey to assess the preference for a new coffee blend among 500 participants. She finds that 89% of the respondents favour the new blend, and she calculates a 95% confidence interval for this percentage to be [77%, 92%]. This information implies that:
- a. There is a probability of 0.89 that the true preference for the new coffee blend falls within the interval [77%, 92%].
 - b. The survey results are accurate for 95% of the respondents.
 - c. 95% of the confidence intervals constructed in similar ways in hypothetical future samples will contain the true percentage of people who prefer the new blend.
 - d. Anna is 95% certain that the true preference for the new coffee blend lies between 77% and 92% and is concentrated at 89%.
- (D)

2. Swathi is a chef and is trying to know about her customers' satisfaction with her soup recipe. She asks customers to rate their satisfaction level by choosing from the following options: very satisfied, satisfied, neutral, unsatisfied, and very unsatisfied. What type of variable best captures these responses?

- a. Nominal
- b. Ratio
- c. Ordinal

(C)

3. A certain email filtering system has a 95% probability of correctly identifying a spam email and a 10% probability of wrongly marking a legitimate email as spam. It is estimated that 0.5% of the emails received by a company are spam. If the system flags an email, what is the probability that it is indeed a spam email?

- a. 0.0455
- b. 0.455
- c. 0.1
- d. 0.095

(A)

4. In a manufacturing plant, the quality control department measures the length of metal rods produced. The lengths of these rods are supposed to have a mean of 150 cm and a standard deviation of 10 cm but the distribution can be skewed to be on the longer side. To verify that the plant was achieving the target mean length of 150 cm, they took a sample of 50 rods to estimate the mean length and found the mean to be 146 cm.

i) What is the quality control department's best guess about the mean length of all the rods produced by the plant? [1 mark]

- a. 150 cm
- b. 146 cm

11. In a research project examining the consistency of a newly developed survey, two groups of raters independently assessed participants' responses. One group rated the responses on odd-numbered days, while the other group rated the same responses on even-numbered days. The aim was to ensure that the survey produced stable results regardless of the day of assessment. Which type of reliability is this study primarily evaluating?

- ☒ a. Test-retest reliability
- b. Inter-rater reliability
- c. Internal consistency
- d. Parallel form reliability

[A]

Assumption: Based on time reference. Hence reliability should still hold when conducted

12. Beth is conducting a research study to assess the cognitive abilities of a group of students over a semester. The students undergo a series of cognitive tests at the beginning, middle, and end of the semester. Beth notices that the scores on the final test are significantly higher than those on the initial test. What is the potential threat to validity that Beth might encounter?

- a. Maturation effect
- b. Regression to mean
- ☒ c. Testing effect
- d. History effect

[C]

Assumption: Due to giving the test multiple times, based on the feedback Beth gets he has improved his score.

13. Emily is conducting a survey to analyse the preferences of smartphone users. To gather responses, she posts a questionnaire on a technology-focused online forum. However, she notices that most of the respondents are avid tech enthusiasts who frequently participate in discussions on the latest gadgets. What type of bias is likely present in Emily's survey?

- a. Non-response bias
- ☒ b. Selection bias
- c. Experimenter bias
- d. Belief bias

[B]

Assumption:

The group of people who mostly participated are tech enthusiasts, hence, a particular section of group. Hence, it's a selection bias. A particular section of group is being selected.

8. Which of the following statements is most accurate about confounding variables in experimental research?

- a. They improve the internal validity of the study
- b. They are variables that the experimenter manipulates directly
- ☒ c. They are related to both predictors and outcomes in a systematic way, posing a threat to internal validity
- d. They are only relevant in observational studies, not experimental research

C

9. In a recent study, researchers evaluated the effectiveness of a new fitness program by gathering participant feedback. The participants were instructed to rate their contentment with the program using a structured system, where the numbers 1 to 10 were utilized, each signifying a consistent and uniform progression in satisfaction levels. Considering the method used for these ratings, identify the most suitable scale of measurement for the data acquired in this study:

- a. Nominal scale
- b. Ordinal scale
- ☒ c. Ratio scale
- d. Interval scale

C

10. Select the correct combination(s) regarding unbiased estimators:

1. An unbiased estimator is utilized for determining the mean of a sample, which is gathered in a random and unbiased manner.
2. An unbiased estimator aims to predict the population parameter without any inherent tendency to either overestimate or underestimate the value of that parameter.
3. When an unbiased estimator is applied to numerous biased samples, it will generally converge towards the population parameter.
4. The standard deviation serves as an example of an unbiased estimator.

A) Only 1 and 4 are correct.

☒ B) Only 2 is correct.

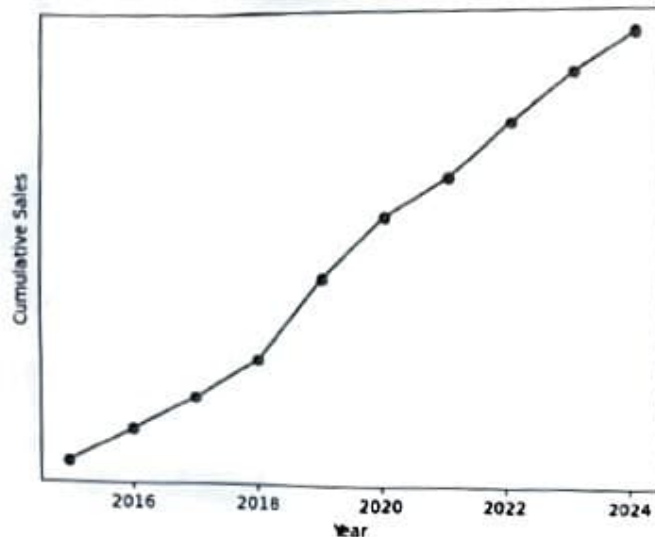
☒ C) Only 2 and 3 are correct.

D) Only 1, 3, and 4 are correct.

E) All 1, 2, 3, and 4 are correct.

C

Q2. A salesman used the below visualization to show the consistent boost in sales. Comment on the visualization used here and why it could be potentially misleading. (2 marks)



Ans This is misleading as the y-axis has cumulative sales & cumulative freq. always increases.

Cumulative sales ~~are~~ at any year means sales upto that year, (i.e. it adds the prev. year sales also)

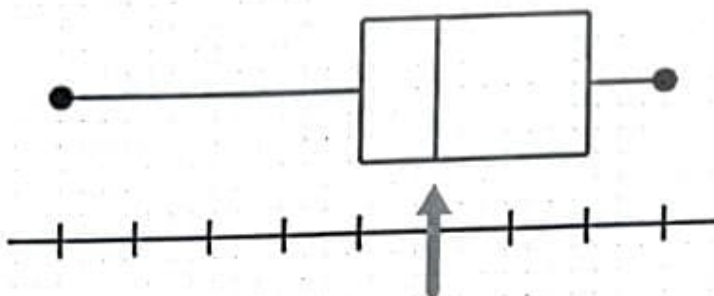
Hence, it's misleading as sales may even drop in consecutive years but cumulative sales never drops.

Layman cannot differentiate it & it thinks it's sales are increasing year-on-year.



Short answers and multi-part questions (12.5 marks):

Q1. Examine the boxplot below and answer the following questions



i) What does the line indicated above by the red arrow signify. (0.5 mark)

- a. Mean
- b. Median
- c. First Quartile
- d. Third Quartile

☒ b

ii) How would you describe the distribution depicted by the boxplot? (0.5 mark)

- a. Uniform Distribution
- b. Left-skewed distribution
- c. Right-skewed distribution
- d. Gaussian distribution

☒ c

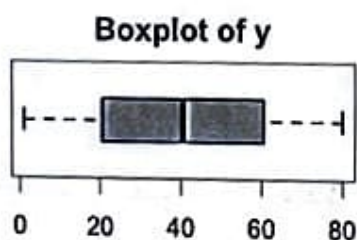
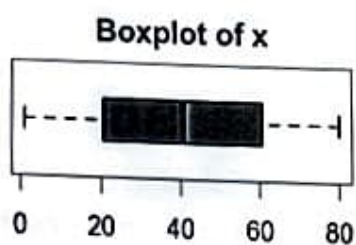
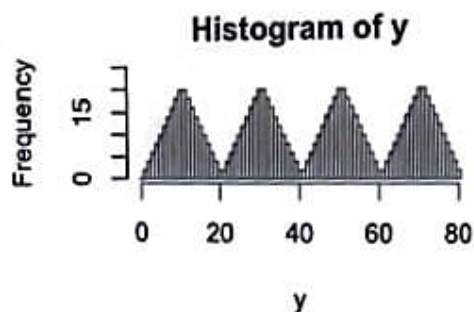
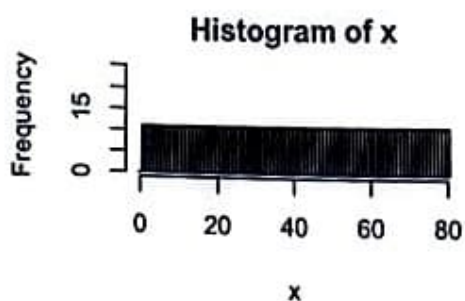
iii) Would you expect the mean and median to align for this distribution: YES / NO (0.5 mark)

YES

iv) Short answer one: Can you think of a real-life example whose measurements will result in the distribution depicted by the boxplot? (1 marks)

eg when you select of ^{sample} ~~group~~ of students from a class.
In that sample if 90% of them are not smart
in terms of marks then you will get this type of
boxplot where it is skewed

(2 marks)



Ans

Boxplot ~~of~~ graph ~~is~~ values remains same.

But histogram graph values & distribution changes.

The reason is as follows:

→ In case of Boxplot:

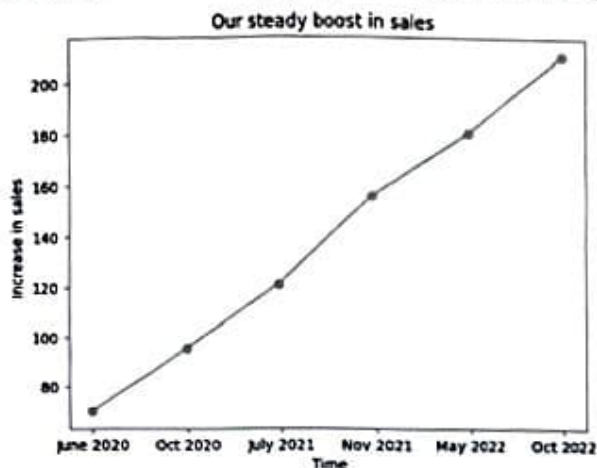
The distribution is based on minimum, first quartile, median, third quartile & maximum values which is same in both ~~of~~ x & y & hence Boxplot graph is same for both of them.

→ In case of Histogram:

It depends upon the Frequency distribution & it is different for x & y & hence histogram graphs are different.

2

Q3. Why is the following graph deceptive? (1 mark)



Ans

~~Here instead of mentioning the m~~
In x-axis the difference in the consecutive values
is not consistent. Hence this way it is
misleading.

②

Q4. Two groups of data (x and y) have been visualized as a boxplot and histogram. As can be seen, while the histograms show a different distribution, the boxplots are the same. Explain why.

Q6: A group of researchers is investigating the average response time of a new software application. They conduct a study on a small sample of participants and calculate a 95% confidence interval for the mean response time to be [50 ms, 70 ms]. Another group of researchers within the same company did a separate study and calculated a 95% CI of [55 ms, 65 ms]. Which group's estimate of average response time from their respective samples is a better guess of the end users' average response time and why? (2 marks)

- Ans
- ~~Q6~~ a Second group's estimate of avg response time ~~is better~~ from their respective samples is a better guess. Because the range of mean response time is smaller in second case. In second case the difference (65-55) is just 10 & in first case it is (70-50)=20.
- When you have a smaller range of ~~a~~ mean response time, it is considered to be better.
 - The range is typically be calculated using this formula.

~~$\mu \pm SEM$~~

$$\mu - (1.96 \times SEM) \leq \bar{x} \leq \mu + (1.96 \times SEM)$$

- If smaller range means smaller value of SEM
 & hence it is better.

~~Not to mention the 1.5~~

The replication crisis in psychology refers to the fact that many published findings cannot be reproduced in later studies. It is caused by issues such as small samples, publication bias, and questionable research practices. It is being addressed through pre-registration, open science, data sharing, and more replication studies.

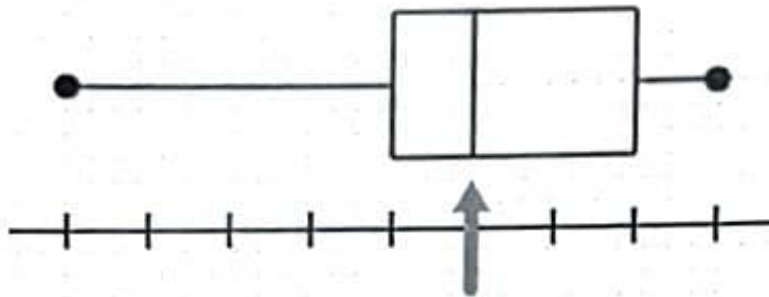
Q5: What is termed "the replication crisis in Psychology"? (1 mark) What were the underlying reasons for it? (1 mark) What are some ways in which we have started addressing those issues? (1 mark)

Ans

- when you do some experiment & when you don't find any significant ~~app~~ outcome on your research, so you keep on modifying ~~the~~ your prior belief & knowledge which you ~~is~~ used in your research ~~do~~ just to give some evident outcome is replication crisis in Psychology.
- when you can't find ~~any~~ outcome to your research you do ~~it~~ this.
- Now as people have become more cautious about the ~~if~~ also & not just the outcomes of the ~~from~~ research & have been interested in the process to outcome also. This way we have started addressing the issue.

Short answers and multi-part questions (12.5 marks):

Q1. Examine the boxplot below and answer the following questions



i) What does the line indicated above by the red arrow signify. (0.5 mark)

- a. Mean
- b. Median
- c. First Quartile
- d. Third Quartile

A

ii) How would you describe the distribution depicted by the boxplot? (0.5 mark)

- a. Uniform Distribution
- b. Left-skewed distribution
- c. Right-skewed distribution
- d. Gaussian distribution

C

iii) Would you expect the mean and median to align for this distribution: YES / NO (0.5 mark)

YES

iv) Short answer one: Can you think of a real-life example whose measurements will result in the distribution depicted by the boxplot? (1 marks)

eg. When you select a sample of students from a class. In that sample if 90% of them are smart in terms of marks then you will get this type of boxplot where it is skewed.

Age at retirement in a company, where most people retire around a common age, but a few retire much earlier.



Information Security Audit and Assurance

Mid Semester Test

28/02/2025

Max Marks: 30

Duration: 90 Min

Answer the following questions. Each question carries 10 marks.

- 1) What is a network firewall? How does it protect a network? Give example of a commercial or open source firewall product, describe its features, advantages, limitations, etc.
- 2) What are VLANs? What are their purposes? How do they work? Discuss different types of VLANs. How VLANs are different from VPNs?
- 3) What are various fundamental rights provisioned in the constitution of Bharat? An important aspect of them is freedom of speech/expressions and privacy. As a person responsible for information security of IT systems, how do you view laws related to freedom of speech/expressions and privacy, how will you ensure them in your organisation, what are their implications to the IT systems. You may take an example IT system or an organisation to explain your views.

Information Security Audit and Assurance
Mid Semester Test
27/02/2024

Max Marks: 20

Duration: 90 Min

Answer one of the following two questions:

1. Do Risk Assessment of an educational institute which is similar to IIT Hyderabad in vision, objectives and activities, and located in Tel Aviv, Israel. Suggest controls to mitigate risks identified. State all assumptions made. Make use of all the concepts that we have discussed in this course so far, your general knowledge and life experiences. Attack
2. Share your understanding of the Aadhaar case in the Supreme Court of India and the final verdict. You may cover the following aspects in your response:
 - a) What were the issues and concerns raised by the petitioners?
 - b) What was the general understanding of the SC Bench?
 - c) What was the outcome/judgement in this case?
 - d) What are your views on the final judgement?
 - e) What are the implications of the judgement?

Threat
Threat Agent
vulnerability
Likelihood
Impact
Risk Mitigation,